MIL-S-13943B
24 March 1983
SUPERSEDING
MIL-S-13943A
2 January 1968

MILITARY SPECIFICATION

SODIUM SULFITE, ANHYDROUS, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers technical grade anhydrous sodium sulfite.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Chemical characteristics. Sodium sulfite shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as Na_2SO_3)	93.0	---	4.2.4.1
Insoluble matter	---	0.2	4.2.4.2
Sodium chloride (NaCl)	---	1.0	4.2.4.3
Iron (Fe)	---	0.1	4.2.4.4
Water content	---	0.5	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sodium sulfite produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.5). Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 120 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

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TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Cushioning missing or not as specified (where required)
(f) Fiberboard pads and partitions missing or not as specified (where required)
(g) Unitization not as specified
(h) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.5). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as Na_2SO_3). Weigh to the nearest milligram (mg) approximately 0.25 g of the specimen and add it to 50.0 milliliters (mL) of 0.1N iodine solution (the sulfite must be added to the iodine solution). Allow to stand for 5 minutes, add 1 mL of hydrochloric acid, and titrate the excess iodine with 0.1N sodium thiosulfate solution using starch as the indicator towards the end of the titration. Calculate the percent by weight sodium sulfite as follows:

$$\text{Percent sodium sulfite} = \frac{6.3\ (AB - CD)}{W}$$

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where: A = Milliliters of iodine solution added,
B = Normality of iodine solution,
C = Milliliters of sodium thiosulfate solution used,
D = Normality of sodium thiosulfate solution, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter. Weigh to the nearest milligram approximately 10 g of the specimen and dissolve in 100 mL of water. Heat to boiling and digest in a covered beaker on a steam bath for 1 hour. Filter through a tared filter crucible of known weight, wash thoroughly with hot water, dry for one hour at 105° + 2°C, cool, and weigh. Calculate the percent by weight insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight of crucible and insoluble matter, in grams,
B = Weight of crucible, in grams, and
W = Weight of specimen in grams.

4.2.4.3 Sodium chloride.

(a) Control solution. Weigh 0.100 g of sodium chloride and dissolve in 1 liter of water.

(b) Procedure. Weigh 1.000 g of the specimen and dissolve in 10 mL of water. Add 3 mL of 30-percent hydrogen peroxide and allow the solution to stand for 30 minutes. Dilute with water to 100 mL. To 50 mL of the solution, add 2 mL of nitric acid and 1 mL of 3-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced in an equal volume of the control solution to which 2 mL of nitric acid and 1 mL of 3-percent silver nitrate solution have been added.

4.2.4.4 Iron.

(a) Control solution. Weigh 0.702 g of ferrous ammonium sulfate, hexahydrate and dissolve in 10 mL of 10-percent sulfuric acid. Dilute with water to 100 mL. To 10 mL of the solution, add 10 mL of 10-percent sulfuric acid and dilute to 1 liter.

(b) Procedure. Weigh 0.100 g of the specimen and dissolve in 10 mL of water. Dilute with water to 100 mL. To 10 mL of the solution, add 2 mL of hydrochloric acid and evaporate to dryness on a steam bath. Dissolve the residue in 5 mL of water and 2 mL of hydrochloric acid and re-evaporate to dryness. Dissolve the residue in 10 mL of water, add 2 mL of hydrochloric acid, dilute to 50 mL with water, add 30 to 50 mg of ammonium persulfate crystals and 3 mL of 30-percent ammonium thiocyanate solution. Any red color produced shall be no darker than that produced by 1 mL of control solution in the same final volume and containing the quantities of reagents used in the specimen test.

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4.2.4.5 Water content. Weigh to the nearest 0.01 g approximately 2 g of the specimen and quantitatively transfer to a crucible. Weigh the crucible and contents to the nearest 0.1 mg. Heat at 105° to 110°C for 2 hours, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent by weight water content as follows:

$$\text{Percent water content} = \frac{100\ (A - B)}{W}$$

where: A = Weight of crucible and specimen before heating, in grams,
B = Weight of crucible and specimen after heating, in grams, and
W = Weight of specimen in grams.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or industrial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packs. Sodium sulfite shall be unit packed in a 1- or 100-pound (lb) quantity as specified (see 6.2).

(a) One-lb quantity. A quantity of 1 lb (+0.10 or -0 ounce) of sodium sulfite shall be unit packed level A in a transparent bag in accordance with submethod IC-3 of MIL-P-116, except that no preservative shall be used, and air need not be completely removed from the bag.

(b) One hundred-lb quantity. A quantity of 100 (+1 or -0) lb of sodium sulfite shall be unit packed in a removable head, gasketed, steel drum furnished with a polyethylene bag liner. The bag liner shall conform to type I, class B, style 1 or 2 of MIL-B-117. The drum shall conform to the requirements of Section 5, Rule 40 of the Uniform Freight Classification (UFC), but shall be cleaned and coated as specified for the drums of PPP-D-729. The bag liner shall be closed by heat sealing, tying, or knotting. The drum shall be closed by bolted locking ring, using a lock-nut or lock washer.

5.1.1.2 Intermediate pack. Twelve bags of sodium sulfite, unit packed as specified in 5.1.1.1(a), shall be intermediately packed in a close-fitting fiberboard box conforming to grade W6c of PPP-B-636. Bags shall be arranged upright in a three by four bag pattern. Motion of contents shall be prevented by insertion of cushioning conforming to PPP-C-843. The box shall be closed as specified for boxes to be overpacked in PPP-B-636.

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5.1.2 Level B.

5.1.2.1 One hundred-lb quantity. A quantity of 100 (+1 or -0) lb of sodium sulfite shall be unit packed level B in a fiber drum with a polyethylene bag liner or in a multiwall paper bag as specified (see 6.2). The drum shall conform to Section 2, Rule 51 of the UFC and shall be furnished with a bag liner as required for the 100-lb quantity of 5.1.1.1(b). The bag shall conform to Section 10, Rule 40 of the UFC and shall be a wet-strength plain paper kraft shipping bag having a minimum 0.001-inch thick polyethylene film innermost wall. The film shall conform to type I or II, class, grade and finish optional of L-P-378. The drum or bag shall be closed to prevent leakage of contents in transit and storage.

5.1.3 Industrial. A 1- or 100-lb quantity of sodium sulfite, as specified (see 6.2), shall be unit packed industrially in accordance with MIL-STD-1188.

5.2 Packing. The preserved sodium sulfite shall be packed level A, B, or industrial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-lb quantity. Four boxes of sodium sulfite, intermediately packed as specified in 5.1.1.2, shall be packed in a close-fitting wood box, cleated plywood box, or wirebound wood box at the option of the contractor. The wood box shall conform to class 2, style 4 of PPP-B-621; the cleated plywood box shall conform to overseas type, style A or B of PPP-B-601; and the wirebound wood box shall conform to class 3, style optional of PPP-B-585. Each box shall be for a type 2 average load. Motion of contents shall be prevented by insertion of fiberboard pads, as needed, made from material conforming to grade W6c of PPP-F-320. Each box shall be closed and reinforced as specified in the applicable box specification.

5.2.1.2 One hundred-lb quantity. The 100-lb quantity, unit packed level A as specified in 5.1.1.1(b), shall require no further protection for shipment other than unitization.

5.2.2 Level B.

5.2.2.1 One-lb quantity. Four boxes of sodium sulfite, intermediately packed as specified in 5.1.1.2, shall be packed in a close-fitting fiberboard box conforming to grade V3c of PPP-B-636, and in a manner as specified for the level A pack in 5.2.1.1.

5.2.2.2 One hundred-lb quantity. The 100-lb quantity of sodium sulfite, unit packed as specified in 5.1.2.1, shall require no further protection for shipment other than unitization.

5.2.3 Industrial. Sodium sulfite shall be packed industrially in accordance with MIL-STD-1188.

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5.3 Unitization. Uniform quantities of level A and B shipping containers shall be palletized in accordance with the applicable requirements in MIL-STD-147. A fiberboard cap, storage aid 4, shall be placed above the top tier of bags and beneath the strapping. Uniform quantities of industrial packs shall be unitized to conform to UFC rules, National Motor Freight Classification rules, and the rules of any other applicable carrier.

5.4 Marking. Level A and B unit packs, intermediate packs, packs, and unitized loads shall be marked in accordance with MIL-STD-129. Industrial unit packs, packs, and pallet loads shall be marked in accordance with MIL-STD-1188. Each container and pallet load shall be marked to show the lot or batch number and date of pack. In addition, each container and pallet load shall be marked to show the following precautionary marking:

"HAZARD
Toxic by Ingestion

WARNING: HARMFUL IF SWALLOWED.
Wash thoroughly after handling.

FIRST AID: If swallowed, induce vomiting by sticking finger down throat or by giving soapy or strong salty water to drink. Repeat until vomit is clear. Call a physician. Never give anything by mouth to an unconscious person."

6. NOTES

6.1 Intended use. Sodium sulfite is intended for use in boiler water treatment.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of preservation and packing required (see 5.1 and 5.2)
(c) Unit quantity required (see 5.1.1.1 and 5.1.3)
(d) Container required for level B unit pack (see 5.1.2.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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6.5 Sampling and testing precautions. This specification covers inspection of chemical material (sodium sulfite) which is potentially hazardous to personnel. All applicable safety rules, regulations and procedures must be followed in the handling and processing of these materials.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD, MI
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Preparing activity:

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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